

Task 1:

1. What are the ranged values for the D1 trace and the D3 trace?

D1: -2.5 to 3 (total range)

X→-1 to 3

Y→-2.5 to 3

Z→-2.5 to 3

D3: -2 to 12 (total range)

X→-2 to 2

Y→2 to 6

Z→6 to 12

2. Why do you think the raw accelerometer data readings look different given that they were collected under similar conditions?

One is a linear accelerometer and the other is a traditional accelerometer. One has the acceleration of gravity included in the movement where the other filters it out and provides the acceleration relative to phone movement rather than all forces acting on the device.

Task 2:

3. Why is it easier to work with the pure magnitude data?

Walking may have a distinct motion in all directions which is amplified when looking at the magnitude. For example, if your phone is in your hand you may swing it with your arm while you walk. Additionally, each step you take may have a swing left to right as your weight shifts. The combination of these three motions can show a larger more subtle movement associated with walking.